Sum-of-Squares degree and size lower bounds for general MCSP

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Editorial note. Selected complete native author main theorem/proof/context, original circuit diagrams and bibliography in standalone layout. Input files explicitly flattened into source spans, unused frontmatter/font packages omitted. No mathematical changes or new proofs. Original CC BY4.0 license retained.

Allowed standard prerequisites (editorial).

- Explicit unbalanced expanders, expander XOR SoS degree lower bound and SoS size-degree tradeoff. Exact cited author statements in preliminaries retained; all local substitution/restriction support supplied in full.

2 Preliminaries

All logarithms are in base 2. For integers $n \geq 1$ we write $[n] = \{1, 2, \ldots, n\}$ and for a set U we denote the power set of U by 2^U. Further, for a set $V \subseteq U$ we let $\overline{V}$ be the complement of V with respect to U, that is, $\overline{V} = U \setminus V$. We write $\binom{[n]}{\ell} \subseteq \{0,1\}^n$ for the set of binary strings with Hamming weight ℓ. For a string $\alpha \in \{0,1\}^n$ we let $|\alpha| = \sum_{i \in [n]} \alpha_i$.

We sometimes want to supress dependencies on constants and write $f(n, \varepsilon) \in O_\varepsilon\big(g(n, \varepsilon)\big)$, respectively $f(n, \varepsilon) \in \Omega_\varepsilon\big(g(n, \varepsilon)\big)$, to mean that there exists a function $c(\varepsilon) > 0$ such that there is an n_0 and for all $n \geq n_0$ it holds that $f(n, \varepsilon) \leq c(\varepsilon) \cdot g(n, \varepsilon)$, respectively $f(n, \varepsilon) \geq c(\varepsilon) \cdot g(n, \varepsilon)$.

Definition 2.1. A sequence of bipartite graphs $\{G_n = (U_n, V_n, E_n)\}_{n \in \mathbb{N}}$ with $\deg(u) = d$ for all $u \in U_n$ is *explicit* if there is an algorithm that given (n, u, j), where $n \in \mathbb{N}, u \in U_n$ and $j \in [d]$, computes the jth neighbor of vertex u in the graph G_n in time $\mathsf{poly}(\log n + \log |U| + \log d)$.

From now on it is understood that whenever we talk about an explicit graph we actually mean to say that there is a sequence of explicit graphs with above properties.

For a graph $G = (V, E)$ and $W \subseteq V$ we denote by

$$N(W) = \{\, u \in V \setminus W \mid (w, u) \in E \text{ for some } w \in W \,\}$$

the set of neighbors of W.

Definition 2.2. A bipartite graph $G = (U, V, E)$ is an (r, d, c)-expander if every vertex $u \in U$ has degree $\deg(u) = d$ and every set $W \subseteq U$ of size $|W| \leq r$ satisfies $|N(W)| \geq c \cdot |W|$.

A key ingredient in our proofs is the following result on the existence of strong explicit expanders.

Theorem 2.3 ([GUV09])**.** *For all constants $\gamma > 0$, every $M \in \mathbb{N}$, $r \leq M$, and $\varepsilon > 0$, there is an $N \leq d^2 \cdot r^{1+\gamma}$ and an explicit $(r, d, (1-\varepsilon)d)$-expander $G = (U, V, E)$, with $|U| = M$, $|V| = N$, and $d = O\big(((\log M)(\log r)/\varepsilon)^{1+1/\gamma}\big)$.*

For our purposes it is more relevant to compute the neighbor relation $\mathbf{Neigh}(u, v)$ indicating whether $(u, v) \in E$ rather than the neighbor function as in Definition 2.1, but this is an immediate consequence of being able to compute the neighbor function.

Claim 2.4. *If $G = (U, V, E)$ is explicit then the neighbor relation $\boldsymbol{Neigh} : U \times V \to \{0, 1\}$ is computable by a circuit of size $d \cdot \big(\mathsf{poly}(\log n + \log |U| + \log d) + 2 \log |V| + 1\big)$.*

A slice function is a Boolean function f such that there is a $\ell \in [n]$ with $f(\alpha) = 0$ whenever $|\alpha| < \ell$, and $f(\alpha) = 1$ whenever $|\alpha| > \ell$. Note that all slice functions are monotone.

The circuit complexity $\mathcal{C}(f)$ of a Boolean function f is the size of the smallest circuit over the basis $\vee, \wedge$, and $\neg$ (with fan-in 2). Similarly the monotone circuit complexity $\mathcal{C}_{\mathsf{mon}}(f)$ of a monotone Boolean function f is the size of the smallest circuit over the basis $\vee$, and $\wedge$. We have the following useful inequality between these measures.

Lemma 2.5 ([Ber82])**.** *If g is any slice function on n bits, then $\mathcal{C}_{\mathsf{mon}}(g) \leq 2\,\mathcal{C}(g) + O(n^2 \log n)$.*

Finally we also rely on the following simple claim.

Claim 2.6. *Let $p : \mathbb{R}^n \to \mathbb{R}$ be a degree d polynomial such that $p(x) = 0$ for all $x \in \{0, 1\}^n$. Then p can be written as*

$$p(x) = \sum_{i \in [n]} q_i(x) \cdot (x_i^2 - x_i)$$

where each term in the sum has degree at most d.

Proof sketch. We take the polynomial p and multilinearize it, using the appropriate polynomial $x_i^2 - x_i$. Eventually we are left with a sum of polynomials of the form $q_i(x) \cdot (x_i^2 - x_i)$ and a multilinear polynomial $\tilde{p}(x)$ which is 0 on all Boolean inputs. As multilinear polynomials are a basis for Boolean functions this implies that $\tilde{p}(x)$ is equal to the 0 polynomial and hence the claim follows. □

2.1 Sum of Squares

Let $\mathcal{P} = \{p_1 = 0, \ldots, p_m = 0\}$ be a system of polynomial equations over the set of variables $X = \{x_1, \ldots, x_n, \bar{x}_1, \ldots, \bar{x}_n\}$. Each p_i is called an *axiom*, and

throughout the paper we always assume that $\mathcal{P}$ includes all axioms $x_i^2 - x_i$ and $\bar{x}_i^2 - \bar{x}_i$, ensuring that the variables are Boolean, as well as the axioms $1 - x_i - \bar{x}_i$, making sure that the "bar" variables are in fact the negation of the "non-bar" variables.

Definition 2.7 (Sum-of-Squares)**.** *Sum-of-Squares (SoS)* is a static, semi-algebraic proof system. An SoS proof of $f \geq 0$ from $\mathcal{P}$ is a sequence of polynomials $\pi = (t_1, \ldots, t_m; s_1, \ldots, s_a)$ such that

$$\sum_{i \in [m]} t_i p_i + \sum_{i \in [a]} s_i^2 = f \ .$$

The *degree* of a proof π is $\mathsf{Deg}(\pi) = \max\{\max_{i \in [m]} \deg(t_i) + \deg(p_i), \max_{i \in [a]} 2\deg(s_i)\}$, an *SoS refutation of* $\mathcal{P}$ is an SoS proof of $-1 \geq 0$ from $\mathcal{P}$, and the *SoS degree to refute* $\mathcal{P}$ is the minimum degree of any SoS refutation of $\mathcal{P}$: if we let π range over all SoS refutations of $\mathcal{P}$, we can write $\mathsf{Deg}(\mathcal{P} \vdash_{\text{SoS}} \perp) = \min_\pi \mathsf{Deg}(\pi)$. The *size of an SoS refutation* π, $\mathsf{Size}(\pi)$, is the sum of the number of monomials in each polynomial in π and the *size of refuting* $\mathcal{P}$ is the minimum size over all refutations $\mathsf{Size}(\mathcal{P} \vdash_{\text{SoS}} \perp) = \min_\pi \mathsf{Size}(\pi)$.

Let us recall some well-known results about SoS. Given a bipartite graph $G = (U, V, E)$, and $b \in \{0,1\}^{|U|}$ we denote by $\Phi(G, b)$ the following XOR-CSP instance defined over G: for each $v \in V$ there is a Boolean variable x_v, and for every vertex $u \in U$ there is a constraint $\oplus_{v \in N(u)} x_v = b_u$. We encode this in the obvious way as a system of polynomial equations:

$$\Big\{ \prod_{v \in N(u)} (1 - 2 \cdot x_v) = 1 - 2 \cdot b_u \mid u \in U \Big\} \ ,$$

along with the Boolean axioms and the negation axioms for the x variables. The first theorem we need to recall is the classic lower bounds for XOR-CSPs by Grigoriev.

Theorem 2.8 ([Gri01])**.** *For $n \in \mathbb{N}$, all $k = k(n)$ and $r = r(n)$ the following holds. Let $G = (U, V, E)$ be an $(r, k, 2)$-expander with $|V| = n$. Then for every $b \in \{0,1\}^{|U|}$ SoS requires degree $\Omega(r)$ to refute the claim that there is a satisfying assignment to $\Phi(G, b)$.*

We also need to recall the size-degree tradeoff by Atserias and Hakoniemi.

Theorem 2.9 ([AH19])**.** *Let $\mathcal{P}$ be a system of polynomial equations over n Boolean variables and degree at most k. If d is the minimum degree SoS requires to refute $\mathcal{P}$, then the minimum size of an SoS refutation of $\mathcal{P}$ is at least $\exp(\Omega((d-k)^2/n))$.*

2.2 Restrictions

Let $\mathcal{P} = \{p_1 = 0, \ldots, p_m = 0\}$ be a system of polynomial equations over the set of Boolean variables $X = \{x_1, \ldots, x_n, \bar{x}_1, \ldots, \bar{x}_n\}$. For a map $\rho : \{x_1, \ldots, x_n\} \to \{0, 1, x_1, \ldots, x_n, \bar{x}_1, \ldots, \bar{x}_n\}$ denote by $\mathcal{P}\big|_\rho$ the system of polynomial equations

$\mathcal{P}$ restricted by ρ, i.e.,

$$\begin{aligned}\mathcal{P}\big|_\rho = \{&p_1(\rho(x_1),\dots,\rho(x_n)) = 0,\\ &p_2(\rho(x_1),\dots,\rho(x_n)) = 0,\\ &\vdots\\ &p_m(\rho(x_1),\dots,\rho(x_n)) = 0\}\ ,\end{aligned}$$

where it is understood that $\rho(\bar{x}_i) = \overline{\rho(x_i)}$, with the convention $\bar{\bar{x}}_i = x_i$, $\bar{0} = 1$ and vice versa. Throughout the paper all our restrictions set the bar variables to the negation of the non-bar variables. As such it makes sense to treat the pair of variables $(x_i, \bar{x}_i)$ as one variable and we say that $\mathcal{P}$ has n *unset* variables.

Definition 2.10 (Variable Substitution)**.** We say that a system of polynomial equations $\mathcal{P}'$ is a *variable substitution of* $\mathcal{P}$ if there is a map $\rho : \{x_1,\dots,x_n\} \to \{0,1,x_1,\dots,x_n,\bar{x}_1,\dots,\bar{x}_n\}$ such that $\mathcal{P}' = \mathcal{P}\big|_\rho$, where we ignore polynomial equations of the form $0 = 0$.

The following well-known lemma states that a system of polynomial equations $\mathcal{P}$ is at least as hard as any of its variable substitutions.

Lemma 2.11. *Let $\mathcal{P}, \mathcal{P}'$ be systems of polynomial equations such that $\mathcal{P}'$ is a variable substitution of $\mathcal{P}$. Then,*

(i) $\mathsf{Deg}(\mathcal{P} \vdash_{SoS} \bot) \geq \mathsf{Deg}(\mathcal{P}' \vdash_{SoS} \bot)$*, and*

(ii) $\mathsf{Size}(\mathcal{P} \vdash_{SoS} \bot) \geq \mathsf{Size}(\mathcal{P}' \vdash_{SoS} \bot)$.

The lemma is easy to verify by considering an SoS refutation of $\mathcal{P}$ and hitting it with the appropriate variable substitution. The restricted proof is now a refutation of $\mathcal{P}'$ and it can be seen that the degree/size of the restricted refutation is at most the degree/size of the original refutation.

We also consider more general substitutions.

Definition 2.12 (Polynomial Substitution)**.** Functions $\rho : \{x_1,\dots,x_n\} \to \mathbb{R}[x]_{\leq k}$ that map variables to polynomials of degree at most k are called *polynomial substitutions*.

For polynomial substitutions we have the following well-known lemma.

Lemma 2.13. *Let $\mathcal{P}$ be a system of polynomial equations and let ρ be a polynomial substitution mapping variables to polynomials of degree at most k. Then,* $\mathsf{Deg}(\mathcal{P} \vdash_{SoS} \bot) \geq \mathsf{Deg}(\mathcal{P}\big|_\rho \vdash_{SoS} \bot)/k$.

This lemma can again be verified by considering a refutation of $\mathcal{P}$. Substitute each variable x_i in the proof by $\rho(x_i)$. This results in a refutation of $\mathcal{P}\big|_\rho$, whose degree is at most a factor k larger than the degree of the refutation of $\mathcal{P}$.

2.3 The Circuit Size Formula

The formula $\mathrm{Circuit}_s(f)$ encodes the claim that the function f, given as a truthtable $f \in \{0,1\}^{2^n}$, can be computed by a circuit of size s over n Boolean inputs $x_1,\dots,x_n$. The encoding is not essential but for concreteness let us fix

one encoding of this claim. We deviate from the encoding used by Razborov [Raz98, Raz04b] and do not present the formula as a propositional formula but rather as a system of polynomial equations. In order to encode below constraints as a constant width CNF formula, as done by Razborov, one needs to introduce extension variables. Despite this difference it is not difficult to see that our lower bound also works against the CNF encoding. In Section 4 we directly show that a low degree SoS refutation of the CNF encoding gives rise to a low degree SoS refutation of the encoding used in this paper (see Proposition 4.1). Thus a lower bound for our encoding implies a lower bound for the CNF encoding. As the presentation is simpler in the polynomial encoding, we present it as follows.

We also need to define the monotone version of $\mathrm{Circuit}_s(f)$ denoted by $\mathrm{Circuit}_s^{\mathsf{mon}}(f)$. The later is a restriction of the former with the $\mathsf{IsNeg}(v)$ (see below) variable, for all $v \in [s]$, set to 0. This forces the circuit to only contain $\wedge$ and $\vee$ gates, i.e., the circuit is monotone.

All variables introduced in the following are Boolean variables and we implicitly add the Boolean axiom $y(1-y) = 0$ for each variable y and further implicitly introduce the "bar variable" $\bar{y}$ along with the negation axiom $y = 1 - \bar{y}$ (and the corresponding Boolean axiom) ensuring that $\bar{y}$ is always the negation of y.

Let us first describe the *structure variables* which are used to describe the circuit that supposedly computes the function f.

We view the s gates as being indexed from 1 to s in topological order with gate s being the output. For each gate $v \in [s]$ there are three variables $\mathsf{IsNeg}(v), \mathsf{IsOr}(v), \mathsf{IsAnd}(v)$ indicating the operation computed at v. Similarly for a gate $v \in [s]$ and a wire $a \in \{1, 2\}$ we have variables $\mathsf{IsFromConst}(v, a), \mathsf{IsFromInput}(v, a), \mathsf{IsFromGate}(v, a)$ indicating whether the input wire a of v is connected to a constant, a variable or a gate.

Further, we have the variables $\mathsf{ConstantValue}(v, a)$, $\mathsf{IsInput}(v, a, i)$ and $\mathsf{IsGate}(v, a, u)$, for $a \in \{1, 2\}$, $i \in [n]$ and $u < v$, specifying the constant value, input x_i or gate u, the input wire a of v is connected to (assuming a is connected to the corresponding kind).

The second set of variables are the *evaluation variables*, which describe what value is computed at each v on input $\alpha = \alpha_1, \ldots, \alpha_n$ (i.e., we have $x_i = \alpha_i$).

For each gate $v \in [s]$ and assignment $\alpha \in \{0, 1\}^n$ we have a Boolean variable $\mathsf{Out}_\alpha(v)$ indicating the value computed at gate v on input α. The Boolean variable $\mathsf{In}_\alpha(v, a)$ indicates the value brought to the vertex $v \in [s]$ on wire $a \in \{1, 2\}$ on input α.

Note that there is a total of $3s + 6s + 2s + 2sn + 2\binom{s}{2} = \Theta(s^2 + sn)$ structure variables, and a total of $3s2^n$ evaluation variables, for a total of $\Theta(s^2 + s2^n)$ variables in $\mathrm{Circuit}_s(f)$.

The formula consists of the following axioms. For the sake of readability we omit some universal quantifiers: the variable $a \in \{1, 2\}$ in Equations (1) and (3) to (9) as well as the variable $\alpha \in \{0, 1\}^n$ in Equations (7) to (13) are implicitly universally quantified.

Let us first describe the axioms on the structure of the circuit. In the following section we refer to this set of axioms as the *structure axioms*. The first axioms ensure that every wire is connected to a single kind

$$\mathsf{IsFromConst}(v, a) + \mathsf{IsFromInput}(v, a) + \mathsf{IsFromGate}(v, a) = 1 \quad \forall\, v \in [s] \ , \quad (1)$$

and similarly the next axioms make sure that each gate is of precisely one kind

$$\mathsf{IsNeg}(v) + \mathsf{IsOr}(v) + \mathsf{IsAnd}(v) = 1 \quad \forall\, v \in [s] \ . \tag{2}$$

The final structure axioms ensure that the variables, which indicate to what input or gate a fixed wire is connected to, always sum to one (except for gate 1 which cannot have any inputs from other gates)

$$\sum_{i=1}^{n} \mathsf{IsInput}(v,a,i) = 1 \quad \forall v \in [s], \text{ and} \tag{3}$$

$$\sum_{u=1}^{v-1} \mathsf{IsGate}(v,a,u) = 1 \quad \forall v \in [s] \setminus \{1\} \ . \tag{4}$$

We further strengthen our encoding by adding the axioms

$$\mathsf{IsInput}(v,a,i)\,\mathsf{IsInput}(v,a,j) = 0 \quad \forall v \in [s],\ i < j \in [n], \text{ and} \tag{5}$$

$$\mathsf{IsGate}(v,a,u)\,\mathsf{IsGate}(v,a,u') = 0 \quad \forall u < u' < v \in [s] \ . \tag{6}$$

Note that Equations (5) and (6) are implied by Equations (3) and (4). We add these axioms in order to argue that a short refutation of the CNF encoding of this principle leads to a short refutation of the present encoding.

The second group of axioms are the *evaluation axioms* and they ensure that the evaluation variables indeed compute the intended values. We start by making sure that the wires carry the value intended by the structure axioms. If a wire is connected to a constant, then the evaluation variable associated with that wire should always be equal to the constant

$$\mathsf{IsFromConst}(v,a) \cdot \big(\mathsf{In}_\alpha(v,a) - \mathsf{ConstantValue}(v,a)\big) = 0 \ , \tag{7}$$

and similarly in case if a wire is connected to an input or a gate

$$\mathsf{IsFromInput}(v,a) \cdot \mathsf{IsInput}(v,a,i) \cdot \big(\mathsf{In}_\alpha(v,a) - \alpha_i\big) = 0 \ , \tag{8}$$

$$\mathsf{IsFromGate}(v,a) \cdot \mathsf{IsGate}(v,a,u) \cdot \big(\mathsf{In}_\alpha(v,a) - \mathsf{Out}_\alpha(u)\big) = 0 \ . \tag{9}$$

The final set of evaluation axioms makes sure that the output evaluation variable of a gate is correctly related to the input evaluation variables:

$$\mathsf{IsNeg}(v) \cdot \mathsf{Out}_\alpha(v) = \mathsf{IsNeg}(v) \cdot \overline{\mathsf{In}_\alpha(v,1)} \ , \tag{10}$$

$$\mathsf{IsOr}(v) \cdot \mathsf{Out}_\alpha(v) = \mathsf{IsOr}(v) \cdot \big(1 - \overline{\mathsf{In}_\alpha(v,1)} \cdot \overline{\mathsf{In}_\alpha(v,2)}\big) \ , \tag{11}$$

$$\mathsf{IsAnd}(v) \cdot \mathsf{Out}_\alpha(v) = \mathsf{IsAnd}(v) \cdot \mathsf{In}_\alpha(v,1) \cdot \mathsf{In}_\alpha(v,2) \ . \tag{12}$$

Last but not least we have the axioms that ensure that the circuit outputs the function specified by the truthtable

$$\mathsf{Out}_\alpha(s) = f(\alpha) \ . \tag{13}$$

3 On Circuits and Restrictions

Let $G = (U, V, E)$ be a bipartite graph with $U = \{0,1\}^n$ and $V = [m]$. As in the XOR-CSP setup (Section 2.1) we think of vertices in U as constraints and

vertices in V as variables. More specifically, we think of each vertex $\alpha \in U$ as an *xor* constraint over the variables in the neighborhood $\oplus_{i \in N(\alpha)} v_i = b_\alpha$, for a constraint vector $b \in \{0,1\}^U$. Given an assignment $\beta \in \{0,1\}^m$ to the variables V, we let $f_{G,\beta} : U \to \{0,1\}$ be the function defined by $f_{G,\beta}(\alpha) = \oplus_{i \in N(\alpha)} v_i$. In other words, viewing $f_{G,\beta}$ as a vector in $\{0,1\}^U$, it is the unique constraint vector such that the XOR-CSP instance, defined over G, is satisfied by the assignment β. Let us denote the set of all such constraint vectors that give rise to a satisfiable XOR-CSP instance by

$$\mathcal{F}_G = \{f_{G,\beta} \,|\, \beta \in \{0,1\}^m\} \ .$$

In order for SoS to refute an XOR-CSP instance defined over G, it must prove that the given constraint vector is not in the set $\mathcal{F}_G$.

On the other hand in order for SoS to refute the formula $\text{Circuit}_s(f)$ it needs to show that there is no circuit of size at most s computing f. That is, SoS needs to show that f is not in the set

$$\mathcal{C}_\emptyset = \{T : \{0,1\}^n \to \{0,1\} \text{ such that } \text{Circuit}_s(T) \text{ is satisfiable}\} \ .$$

More generally, if we restrict $\text{Circuit}_s(f)$ by a restriction ρ, then the proof system must prove that f is not a member of the family of truthtables

$$\mathcal{C}_\rho = \{T : \{0,1\}^n \to \{0,1\} \text{ such that } \text{Circuit}_s(T)\big|_\rho \text{ is satisfiable}\} \ .$$

In the following we show that there is a well-behaved restriction ρ such that $\mathcal{C}_\rho = \mathcal{F}_G$ for some explicit graphs G. In other words, once we consider the restricted formula $\text{Circuit}_s(f)\big|_\rho$, SoS needs to rule out that f is a valid right hand side of an XOR-CSP instance. But we know that if G is a moderate expander, then low degree SoS cannot determine whether the XOR-CSP instance is satisfiable and hence we obtain our lower bound.

Let us first formalize the properties we require from ρ. We start off by restricting our attention to a certain natural class of variable substitutions. Namely, we do not want that the structure of the circuit depends on evaluation variables.

Definition 3.1 (natural variable substitutions)**.** A variable substitution ρ to the variables of $\text{Circuit}_s(f)$ is *natural* if there is *no* structure variable y such that $\rho(y)$ is an evaluation variable.

In order to motivate the next definition, let us informally describe the natural restriction ρ and explain the properties of ρ we require.

For now we can think of ρ as a restriction to the structure variables (though for the size lower bounds we also need to restrict some of the evaluation variables). Some set of m structure variables remains undetermined. Let us denote these variables by $y_1, \ldots, y_m$. We intend to choose ρ such that on a given input $\alpha \in \{0,1\}^n$ to the circuit, it is forced to compute $\oplus_{i \in N(\alpha)} y_i$. In other words, given such a restriction ρ, we are *essentially* left with an XOR-CSP problem over G, with right hand side f. There is however a difference in that the encoding is non-standard: the evaluation variables act like extension variables that correspond to the functions computed at each gate of the circuit. In order to argue that the known degree lower bound for the XOR-CSP problem implies a

degree lower bound for the problem at hand, we need to get rid of these extension variables. This can be done if the functions computed at the gates are of low degree in the y variables.

Recall from Section 2.2 that a system of polynomial equations $\mathcal{P}$ has n unset variables if there are n tuples of variables $(x, \bar{x})$ such that at least one variable of each tuple occurs in $\mathcal{P}$ and all variables in these tuples are unset, i.e., they are not fixed to a constant.

Definition 3.2 (k-determined)**.** Let ρ be a variable substitution to the variables of $\text{Circuit}_s(f)$ and suppose that ρ leaves m structural variables $Y = \{y_1, \ldots, y_m\}$ unset. Then ρ is *k-determined* if for every $v \in [s]$ and $\alpha \in \{0,1\}^n$ there are multilinear polynomials

$$g^{\text{out}}_{v,\alpha}, g^{\text{in}_1}_{v,\alpha}, g^{\text{in}_2}_{v,\alpha} : \{0,1\}^m \to \{0,1\}$$

depending on at most k variables such that the following holds. For all $T \in \mathcal{C}_\rho$ and all total assignments σ that satisfy $\text{Circuit}_s(T)\big|_\rho$ it holds that

$$\mathsf{Out}_\alpha(v)\big|_{\rho\cup\sigma} = g^{\text{out}}_{v,\alpha}(\beta) \ , \quad \mathsf{In}_\alpha(v,1)\big|_{\rho\cup\sigma} = g^{\text{in}_1}_{v,\alpha}(\beta) \ , \text{ and } \quad \mathsf{In}_\alpha(v,2)\big|_{\rho\cup\sigma} = g^{\text{in}_2}_{v,\alpha}(\beta) \ , \tag{14}$$

where $\beta \subseteq \sigma$ is the assignment to Y.

However, Definition 3.2 is not quite sufficient. For example, there is no guarantee that $\mathcal{C}_\rho$ is non-empty, i.e., that the restriction ρ describes a valid (partial) circuit. More generally, we need the additional guarantee that there are still many viable circuits that the restricted formula can describe: if there is just a single setting of the Y variables such that all structural axioms are satisfied, then the formula may be refuted in constant degree. Hence we need to ensure that there are many viable assignments to the Y variables that satisfy all structure axioms. This leads us to the following definition.

Definition 3.3 (m-independent)**.** A variable substitution ρ to the variables of the formula $\text{Circuit}_s(f)$ is *m-independent* if ρ leaves exactly m structural variables $Y = \{y_1, \ldots, y_m\}$ unset, and for every assignment $\beta \in \{0,1\}^Y$ it holds that $|\mathcal{C}_{\rho\cup\beta}| = 1$.

With these definitions at hand we can state the lemma that drives all our lower bounds.

Lemma 3.4. *Let ρ be a natural m-independent k-determined variable substitution of $\text{Circuit}_s(f)$, and let Y and $g^{\text{out}}_{u,\alpha}$ be as in Definition 3.2. If there is an SoS refutation of $\text{Circuit}_s(f)\big|_\rho$ of degree d, then there is a degree $d \cdot k$ SoS refutation of the system of polynomial equations*

$$\{g^{\text{out}}_{s,\alpha}(Y) = f(\alpha) \mid \alpha \in \{0,1\}^n\} \cup \{y_i^2 = y_i \mid i \in [m]\} \ . \tag{15}$$

For proving this lemma, we consider the natural extension of ρ which substitutes all evaluation variables by appropriate degree-k polynomials as indicated by Definition 3.2.

Definition 3.5. For a k-determined restriction ρ (with associated polynomials $g^{\text{out}}_{v,\alpha}, g^{\text{in}_1}_{v,\alpha}, g^{\text{in}_2}_{v,\alpha}$) of $\text{Circuit}_s(f)$, we denote by $\hat{\rho}$ the polynomial substitution that

extends ρ by first substituting any bar variable $\bar{x}$ by $1-x$ and then substituting all evaluation variables as follows:

$$\hat{\rho}(\mathsf{Out}_\alpha(v)) = g^{\text{out}}_{v,\alpha}(Y) \ , \quad \hat{\rho}(\mathsf{In}_\alpha(v,1)) = g^{\text{in}_1}_{v,\alpha}(Y) \ , \text{ and } \quad \hat{\rho}(\mathsf{In}_\alpha(v,2)) = g^{\text{in}_2}_{v,\alpha}(Y) \ .$$

Note that the formula $\text{Circuit}_s(f)\big|_{\hat{\rho}}$ is defined only over Y. Let us stress that there are no "bar" variables left in the formula. The main observation used to prove Lemma 3.4 is the following claim, which establishes that the formula (15) is in fact essentially the same as $\text{Circuit}_s(f)\big|_{\hat{\rho}}$.

Claim 3.6. *Let ρ be a natural m-independent k-determined variable substitution of* $\text{Circuit}_s(f)$. *Then* $\text{Circuit}_s(f)\big|_{\hat{\rho}}$ *can be written as*

$$\text{Circuit}_s(f)\big|_{\hat{\rho}} = \mathcal{P} \cup \mathcal{Q} \ ,$$

where $\mathcal{P}$ is the formula (15) *and $\mathcal{Q}$ only consists of axioms that are satisfied for all assignments $\beta \in \{0,1\}^Y$.*

Proof. Note that the set of output axioms (13) of $\text{Circuit}_s(f)$ under $\hat{\rho}$ equals the first part of (15), and that the Boolean axioms on the Y variables in $\text{Circuit}_s(f)\big|_{\hat{\rho}}$ are exactly the second part of (15).

The remaining axioms of $\text{Circuit}_s(f)\big|_{\hat{\rho}}$, which are not present in (15), are the Boolean axioms on the variables outside Y, the negation axioms, as well as Equations (1) to (12).

The Boolean axioms may turn into polynomials of degree at most $2k$. Because the polynomials we substitute the variables with are Boolean valued, we see that these substituted axioms are satisfied for all assignments $\beta \in \{0,1\}^Y$ and we can thus put them into the set $\mathcal{Q}$.

The negation axioms all become "$0 = 0$" under $\hat{\rho}$ since $\hat{\rho}(\bar{x}) = 1 - \hat{\rho}(x)$.

Finally we need to argue that the Equations (1) to (12) are also of the form $p(Y) = 0$ for a polynomial p which is identically 0 on all of $\{0,1\}^m$. This in turn follows immediately from the assumption that ρ is m-independent: for every $\beta \in \{0,1\}^Y$, there exists some T such that the complete assignment $\hat{\rho}(\beta) \cup \beta$ satisfies $\text{Circuit}_s(T)$. But since none of the remaining Equations (1) to (12) depends on T, they must then all be satisfied for every $\beta \in \{0,1\}^Y$. □

Using this claim we can easily prove Lemma 3.4.

Proof of Lemma 3.4. Suppose $\text{Circuit}_s(f)\big|_{\rho}$ has a refutation in degree d. By Lemma 2.13, there then exists a degree $d \cdot k$ refutation of $\text{Circuit}_s(f)\big|_{\hat{\rho}}$.

By Claim 3.6, this new refutation is *almost* a refutation of (15), except that $\text{Circuit}_s(f)\big|_{\hat{\rho}}$ has an additional set $\mathcal{Q}$ of axioms that the refutation may use. However, each of these additional axioms is of the form $p(Y) = 0$ for a polynomial which is identically 0 on the entire Boolean cube. By Claim 2.6, such an axiom can be rewritten as a linear combination of the Boolean axioms. Since the Boolean axioms are present in (15), this yields a refutation of that formula in degree $d \cdot k$. □

4 On Encodings of the Circuit$_s(f)$ Tautology

Let us introduce a possible constant width CNF encoding of Circuit$_s(f)$ as proposed by Razborov [Raz98].

The formula is defined over the same variables as introduced in Section 2.3, but in order to keep the fan-in bounded, we further introduce the extension variables $\mathsf{IsInput}^{\leq}(v,a,i)$ and $\mathsf{IsGate}^{\leq}(v,a,u)$ that indicate whether the wire a of v is connected to a variable in $x_1,\ldots,x_i$, a gate $1,\ldots,u$ respectively.

Let us group the axioms in the same manner as we did in Section 2.3. First we have the structure axioms. The first axioms encode that each wire is connected to a single kind

$$\begin{aligned}&\big(\mathsf{IsFromConst}(v,a)\vee\mathsf{IsFromInput}(v,a)\vee\mathsf{IsFromGate}(v,a)\big)\wedge\\&\neg\big(\mathsf{IsFromConst}(v,a)\wedge\mathsf{IsFromInput}(v,a)\big)\wedge\\&\neg\big(\mathsf{IsFromInput}(v,a)\wedge\mathsf{IsFromGate}(v,a)\big)\wedge\\&\neg\big(\mathsf{IsFromConst}(v,a)\wedge\mathsf{IsFromGate}(v,a)\big)\ .\end{aligned}\tag{16}$$

The next set of axioms similarly ensures that each gate computes precisely one function

$$\begin{aligned}&\big(\mathsf{IsNeg}(v)\vee\mathsf{IsOr}(v)\vee\mathsf{IsAnd}(v)\big)\wedge\\&\neg\big(\mathsf{IsNeg}(v)\wedge\mathsf{IsOr}(v)\big)\wedge\neg\big(\mathsf{IsOr}(v)\wedge\mathsf{IsAnd}(v)\big)\wedge\neg\big(\mathsf{IsNeg}(v)\wedge\mathsf{IsAnd}(v)\big)\ .\end{aligned}\tag{17}$$

Last, we need to make sure that each wire is connected to a single input or a gate.

$$\begin{aligned}&\mathsf{IsInput}^{\leq}(v,a,n)\wedge\bigwedge_{i\neq j}\neg\big(\mathsf{IsInput}(v,a,i)\wedge\mathsf{IsInput}(v,a,j)\big)\wedge\\&\bigwedge_{i\in[n]}\Big(\mathsf{IsInput}^{\leq}(v,a,i)\equiv\big(\mathsf{IsInput}^{\leq}(v,a,i-1)\vee\mathsf{IsInput}(v,a,i)\big)\Big)\ ,\\&\qquad\text{where }\ \mathsf{IsInput}^{\leq}(v,a,0)\stackrel{\text{def}}{=}0\ ,\end{aligned}\tag{18}$$

and similarly for $v\in[s]\setminus\{1\}$ we have that

$$\begin{aligned}&\mathsf{IsGate}^{\leq}(v,a,v-1)\wedge\bigwedge_{u<u'<v}\neg\big(\mathsf{IsGate}(v,a,u)\wedge\mathsf{IsGate}(v,a,u')\big)\wedge\\&\bigwedge_{u\in[v-1]}\Big(\mathsf{IsGate}^{\leq}(v,a,u)\equiv\big(\mathsf{IsGate}^{\leq}(v,a,u-1)\vee\mathsf{IsGate}(v,a,u)\big)\Big)\ ,\\&\qquad\text{where }\ \mathsf{IsGate}^{\leq}(v,a,0)\stackrel{\text{def}}{=}0\ .\end{aligned}\tag{19}$$

Let us take a look at the evaluation axioms. Again, we have axioms that ensure that the wires carry the values intended by the structure variables. If a wire is connected to a constant, then the evaluation variable associated with that wire should be equal to the constant

$$\mathsf{IsFromConst}(v,a)\to\big(\mathsf{In}_{\alpha}(v,a)\equiv\mathsf{ConstantValue}(v,a)\big)\ ,\tag{20}$$

and similarly if a wire is connected to an input or a gate

$$\mathsf{IsFromInput}(v,a) \wedge \mathsf{IsInput}(v,a,i) \to \mathsf{In}_\alpha(v,a) \equiv \alpha_i \ , \tag{21}$$

$$\mathsf{IsFromGate}(v,a) \wedge \mathsf{IsGate}(v,a,u) \to \mathsf{In}_\alpha(v,a) \equiv \mathsf{Out}_\alpha(u) \ . \tag{22}$$

Last we need to make sure that the gates propagate the value they are supposed to compute.

$$\mathsf{IsNeg}(v) \to \big(\mathsf{Out}_\alpha(v) \equiv \neg\, \mathsf{In}_\alpha(v,1)\big) \tag{23}$$

$$\mathsf{IsOr}(v) \to \big(\mathsf{Out}_\alpha(v) \equiv \mathsf{In}_\alpha(v,1) \vee \mathsf{In}_\alpha(v,2)\big) \tag{24}$$

$$\mathsf{IsAnd}(v) \to \big(\mathsf{Out}_\alpha(v) \equiv \mathsf{In}_\alpha(v,1) \wedge \mathsf{In}_\alpha(v,2)\big) \ . \tag{25}$$

The final axioms ensure that the correct function is computed

$$\mathsf{Out}_\alpha(s) \equiv f(\alpha) \ . \tag{26}$$

This formula can be rewritten in the usual manner into a 4-CNF. Let us denote this formula by $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)$.

Proposition 4.1. *If there is an SoS refutation of degree d of the CNF formula $\mathrm{Circuit}_s^{CNF}(f)$, then there is an SoS refutation of degree $O(d)$ of the system of polynomials $\mathrm{Circuit}_s(f)$ as introduced in Section 2.3.*

The rest of this section is devoted to the proof of Proposition 4.1.

Observe that for each axiom p from the polynomial encoding $\mathrm{Circuit}_s(f)$, there is a CNF $F_p \subseteq \mathrm{Circuit}_s^{\mathrm{CNF}}(f)$ over the same variables as p (ignoring the added extension variables $\mathsf{IsInput}^{\leq}(v,a,i)$ and $\mathsf{IsGate}^{\leq}(v,a,u)$) such that $p(\alpha) = 0$ is satisfied by a Boolean assignment α if and only if F_p is satisfied by α (where we extend the assignment to the extension variables in the natural manner).

Recall that SoS operates on polynomials and we thus need to translate the CNF into a system of polynomials. We translate a clause $\vee_{i\in[w]} z_i$ into the polynomial $\prod_{i\in[w]}(1-z_i) = 0$.

Observe that almost all axioms p of $\mathrm{Circuit}_s(f)$ depend only on a constant number of variables. From such p, using the appropriate Boolean axioms and negation axioms, we can in constant degree derive F_p.

Let us define a polynomial substitution ρ that gets rid of the extension variables in the natural manner: the substitution ρ first replace each occurrence of a "bar" extension variable $\overline{\mathsf{IsInput}^{\leq}(v,a,i)}$ or $\overline{\mathsf{IsGate}^{\leq}(v,a,u)}$ by the polynomial $1-\mathsf{IsInput}^{\leq}(v,a,i)$ and the polynomial $1-\mathsf{IsGate}^{\leq}(v,a,u)$ respectively. Then, ρ replaces each occurrence of the variable $\mathsf{IsInput}^{\leq}(v,a,i)$ by $\sum_{j\leq i}\mathsf{IsInput}(v,a,j)$ and similarly $\mathsf{IsGate}^{\leq}(v,a,u)$ by $\sum_{w\leq u}\mathsf{IsGate}(v,a,w)$.

Suppose we have a degree d refutation π of $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)$. Let us consider $\pi\big|_\rho$ and $\mathrm{Circuit}^{\mathrm{CNF}}(f)\big|_\rho$. Note that $\pi\big|_\rho$ is a degree d SoS refutation of $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)\big|_\rho$.

We claim that in constant degree the axioms of $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)\big|_\rho$ can be derived from the polynomial encoding $\mathrm{Circuit}_s(f)$. As previously noted, this holds for all axioms but the ones that are over a non-constant number of variables. In other words it just remains to show that we can derive the substituted Equations (18) and (19) from Equations (3) to (6).

Let us consider Equation (18). With the extension variables substituted and translated into a system of polynomials the axiom consists of the following polynomial equations.

$$1 - \sum_{j \in [n]} \mathsf{IsInput}(v, a, j) = 0 \tag{27}$$

$$\mathsf{IsInput}(v, a, i) \cdot \mathsf{IsInput}(v, a, j) = 0, \text{ for } i \neq j \tag{28}$$

$$\left(\sum_{j \leq i} \mathsf{IsInput}(v, a, j)\right) \left(1 - \sum_{j < i} \mathsf{IsInput}(v, a, j)\right) \cdot \overline{\mathsf{IsInput}(v, a, i)} = 0, \text{ for } i \in [n] \tag{29}$$

$$\left(1 - \sum_{j \leq i} \mathsf{IsInput}(v, a, j)\right) \left(\sum_{j < i} \mathsf{IsInput}(v, a, j)\right) = 0, \text{ for } i \in [n] \tag{30}$$

$$\left(1 - \sum_{j \leq i} \mathsf{IsInput}(v, a, j)\right) \mathsf{IsInput}(v, a, i) = 0, \text{ for } i \in [n] \ . \tag{31}$$

Equation (27) is equal to Equation (3) and similarly Equation (28) is equal to Equation (5). In the following we show that Equations (29) to (31) can be derived from Equation (5), the Boolean axioms and the negation axioms in constant degree.

Consider Equation (29). Expand and rewrite modulo the Boolean axioms and the negation axiom to obtain

$$\mathsf{IsInput}(v, a, i) \left(\sum_{j < i} \mathsf{IsInput}(v, a, j)\right)^2 - 2 \sum_{j < j' < i} \mathsf{IsInput}(v, a, j) \cdot \mathsf{IsInput}(v, a, j') - \mathsf{IsInput}(v, a, i) \sum_{j < i} \mathsf{IsInput}(v, a, j) = 0 \ . \tag{32}$$

Observe that every term t left in this polynomial is of the form $t = t' \cdot \mathsf{IsInput}(v, a, j) \cdot \mathsf{IsInput}(v, a, j')$, for some $j \neq j' \in [i]$ and a term t' of degree at most 1. But this means that every term is equal to 0 modulo Equation (5) and we thus see that Equation (29) can be derived in constant degree from $\text{Circuit}_s(f)$.

Let us consider Equation (30). Rewrite modulo the Boolean axiom to obtain

$$\mathsf{IsInput}(v, a, i) \sum_{j < i} \mathsf{IsInput}(v, a, j) + 2 \sum_{j < j' < i} \mathsf{IsInput}(v, a, j) \cdot \mathsf{IsInput}(v, a, j') = 0 \ . \tag{33}$$

All terms are of the form of Equation (5) and we can thus derive Equation (30) from $\text{Circuit}_s(f)$ in constant degree.

Last, we need to consider Equation (31). Note that modulo the Boolean

axiom we obtain the polynomial equation

$$-\mathsf{IsInput}(v,a,i)\sum_{j<i}\mathsf{IsInput}(v,a,j)=0\ . \tag{34}$$

Also in this polynomial every term is of the form of Equation (5) and thus also Equation (31) can be derived in constant degree.

What remains is to show that Equation (19) can be derived from $\mathrm{Circuit}_s(f)$ in constant degree. This can be checked analogous to Equation (18) and we thus omit it here.

We conclude that all axioms of $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)\big|_\rho$ can be derived from $\mathrm{Circuit}_s(f)$ in constant degree and thus a degree d SoS refutation of $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)$ gives rise to a degree $O(d)$ SoS refutation of $\mathrm{Circuit}_s(f)$. Equivalently, a degree d lower bound for $\mathrm{Circuit}_s(f)$ implies a degree $\Omega(d)$ lower bound for $\mathrm{Circuit}_s^{\mathrm{CNF}}(f)$ as claimed.

The lower bound of $\Omega_\varepsilon(s^{1-\varepsilon})$ on the degree is essentially tight: if f does not have a circuit of size s then there exists an SoS refutation of this statement in degree $O(s)$.

Proposition 4.2. *Let $s \in N$ and $f:\{0,1\}^n\to\{0,1\}$ be a Boolean function on n bits that requires circuits of size larger than s to be computed. Then there is a degree $O(s)$ SoS refutation of* $\mathrm{Circuit}_s(f)$.

We also prove a result about the minimum size (number of monomials) required for SoS to refute $\mathrm{Circuit}_s(f)$. This result holds for all functions that "almost" have a circuit of size s, in the sense that they have an errorless heuristic circuit (see the survey [BT06]) of size $s/2$ and extremely small error probability with respect to the uniform distribution. Formally, we let $\mathcal{F}_n(s,t)$ denote the class of Boolean functions that consists of all functions $f:\{0,1\}^n\to\{0,1\}$ for which there is a Boolean circuit $C_f:\{0,1\}^n\to\{0,1,\perp\}$ of size at most s such that

1. if $C_f(\alpha)\neq\perp$, then $C_f(\alpha)=f(\alpha)$, and
2. $C_f(\alpha)=\perp$ on at most t inputs.

In other words the circuit C_f computes f correctly on all except t inputs. Note that technically the output of the circuit C_f is two bits with the first one indicating whether the output is $\perp$ or the value of the second bit. We believe that above presentation is more intuitive and hope that the slight abuse of notation causes no confusion. With the class of functions $\mathcal{F}_n(s,t)$ at hand we can state our main SoS size lower bound.

Theorem 4.3. *For all $\varepsilon>0$ there is a $d=d(\varepsilon)$ such that the following holds. For $n\in\mathbb{N}$, all $s\geq n^d$ and any Boolean function $f:\{0,1\}^n\to\{0,1\}$ on n bits, SoS requires degree $\Omega_\varepsilon(s^{1-\varepsilon})$ to refute* $\mathrm{Circuit}_s(f)$.

Theorem 4.4. *For all $\varepsilon>0$ there is a $d=d(\varepsilon)$ such that the following holds. Let $n\in N$ and $s\in\mathbb{N}$ such that $s\geq n^d$. If $t\geq s$ and $f\in\mathcal{F}_n(s/2,t)$, then it holds that SoS requires size* $\exp\left(\Omega_\varepsilon(s^{2-\varepsilon}/t)\right)$ *to refute* $\mathrm{Circuit}_s(f)$.

5 Lower Bounds for General Circuits

We state the following lemma general enough so that we can apply it for the degree as well as the size lower bound. As explained previously, for the size lower bounds we rely on functions that almost have circuits of size s. Recall that we consider the class of functions $\mathcal{F}_n(s,t)$ that consists of all Boolean functions $f : \{0,1\}^n \to \{0,1\}$ for which there is a Boolean circuit $C_f : \{0,1\}^n \to \{0,1,\perp\}$ of size at most s such that

1. if $C_f(\alpha) \neq \perp$, then $C_f(\alpha) = f(\alpha)$, and
2. $C_f(\alpha) = \perp$ on at most t inputs.

The following lemma establishes the existence of m-independent k-determined variable substitutions that result in XOR-CSP instances over explicit graphs.

Lemma 5.1. *For all $k, m, n, t \in \mathbb{N}$ satisfying $m \leq 2^n$, and any explicit bipartite graph $G = (U, V, E)$ such that $|U| = 2^n$, $|V| = m$ and all $u \in U$ are of degree $\deg(u) \leq k$, the following holds. There is a constant $C > 0$, depending on the explicitness of G, such that for all $s \geq C \cdot m \cdot n^C \cdot k^C$ and any Boolean function $f \in \mathcal{F}_n(s/2, t)$ there is a natural m-independent k-determined variable substitutions ρ for the formula $\mathrm{Circuit}_s(f)$ such that*

$$g^{\mathrm{out}}_{s,\alpha}(Y) = \begin{cases} f(\alpha), & \text{if } C_f(\alpha) \neq \perp, \\ \oplus_{i \in N(\alpha)} y_i, & \text{otherwise} \end{cases}$$

for all $\alpha \in \{0,1\}^n$ and $g^{\mathrm{out}}_{s,\alpha}$ and Y as in Definition 3.2. Furthermore, the formula $\mathrm{Circuit}_s(f)\big|_\rho$ is over $O\big(t \cdot k + m\big)$ variables.

For the degree lower bound (Theorem 4.3) we will set $t = 2^n$ and use the trivial C_f which always outputs $\perp$, so the reader who wishes a simplified version of the lemma can focus on this special case.

Proof. We consider the formula $\mathrm{Circuit}_s(f)$ and let the first m gates of the formula be denoted by Y. We restrict the formula such that each gate in Y computes an *or* of two constants. The first wire to the gate is fixed to the constant 0, whereas the second wire is only restricted to carry either the constant 0 or 1. In the end these will be the only structural variables that are not restricted to a constant. In the following we think of the gates Y as Boolean variables; as m additional input bits to our circuit.

Further, we restrict another part of the formula such that one part of the circuit described by the formula computes the circuit C_f. Recall that we pretend that the output of C_f is in $\{0,1,\perp\}$, but it actually outputs two bits C^1_f and C^2_f, where $C^1_f(\alpha) = 1$ if and only if $C^2_f(\alpha) = f(\alpha)$.

Finally we also want to hard code the bipartite graph $G(\{0,1\}^n, Y, E)$ into our circuit. Since G is very large this requires G to be explicit. That is, we require small circuits $\mathsf{Sel}_1, \ldots, \mathsf{Sel}_m$, where given any $\alpha \in \{0,1\}^n$, $\mathsf{Sel}_i(\alpha)$ is 1 if and only if the vertex $y_i \in Y$ is a neighbor of the vertex α. By Claim 2.4 these circuits Sel_i are each of size

$$k \cdot (\mathsf{poly}(n + \log k) + 2 \log m + 1) \leq \mathsf{poly}(n, k) \ .$$

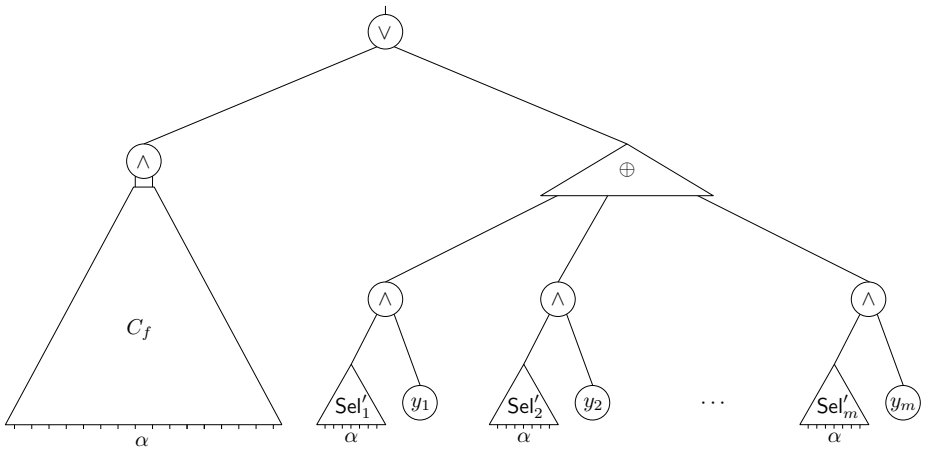


Figure 1: A schematic depiction of the formula after hitting it with the described restriction.

The restriction ρ restricts some structural variables such that a part of the circuit computes $\mathsf{Sel}_1, \ldots, \mathsf{Sel}_m$. We connect each output of the Sel_i circuit by an *and* gate to the negation of C^1_f. Denote the resulting circuits by $\mathsf{Sel}'_1, \ldots, \mathsf{Sel}'_m$. Observe that the circuits Sel'_i output 0 whenever $C^2_f(\alpha) = f(\alpha)$ and otherwise output Sel_i. We think of these circuits as "selector circuits" which indicate whether on input $\alpha \in \{0,1\}^n$ (to the original variables $x_1, \ldots, x_n$ over which the circuit is defined) the variable $y_i \in Y$ appears in the constraint for α.

The output of these selector circuits Sel'_i is connected to the gate y_i by an *and* gate. All these m *and* gates are in turn connected to a circuit computing the *xor* of these gates. Finally, to ensure that the circuit computes $f(\alpha)$ on inputs α such that $C_f(\alpha) \neq \bot$, we connect C^1_f with C^2_f by an *and* gate which is then connceted by a *or* gate to the output of the *xor* circuit. This completes the description of the restriction on the structure variables. A depiction of the resulting circuit can be found in Figure 1.

Note that this implements the intended semantics: for each input $\alpha \in \{0,1\}^n$ the selector circuits output 1 on some variables y_i which are then *xor*'ed, and the restricted circuit outputs

$$\bigoplus_{i \in N(\alpha)} y_i \ , \tag{35}$$

unless $C_f(\alpha) \neq \bot$, in which case the output of the circuit is $f(\alpha)$ and all selector circuits output 0. We require that s is larger than the size of the described circuit which is of size $O\big(m \cdot \mathsf{poly}(n,k)\big) + s/2$.

We have the intended semantics of the circuit and need to ensure the furthermore property: that the restricted formula is over few variables. First, since the selector circuits Sel'_i are fixed, all evaluation variables for these subcircuits can be fixed to constants. The same holds for the circuit C_f. Similarly, since the y_i gate always carries the value of the y_i variable, all $2^n \cdot m$ wire variables corresponding to the Y variables can be substituted by the corresponding y_i variable and are thus restricted away.

After these restrictions the only evaluation variables left are those for the

evaluation of the $\oplus$ circuit. For α such that $C_f(\alpha) \neq \perp$, the selector circuits are hard-wired to 0 and in particular the inputs to the $\oplus$ circuit is hard-wired to 0, meaning that these evalation variables can be restricted away.

There remains then only the $O(t \cdot m)$ evaluation variables corresponding to the evaluation of the $\oplus$ circuit for inputs α such that $C_f(\alpha) = \perp$. Let us, without loss of generality, use an *xor*-circuit which iteratively *xor*s each variable. Concretely, let it have subcircuits χ_i where $\chi_1 = \mathsf{Sel}'_1 \wedge y_1$ and $\chi_i = \chi_{i-1} \oplus (\mathsf{Sel}'_i \wedge y_i)$ for $i > 1$, and χ_m is the overall output of the $\oplus$ circuit.

The only observation required is that if the circuit $\mathsf{Sel}'_i(\alpha) = 0$, then χ_i gets a 0 as input from index i, independent of the value of y_i. Hence the output wire variable of the circuit χ_i indexed by the input α can be substituted by the output of the circuit χ_{i-1}. Hence for each α such that $C_f(\alpha) = \perp$, we can reduce the number of free wire variables indexed by α to $O(k)$, as each $\oplus$-constraint is over at most k variables. As C_f outputs $\perp$ on at most t inputs, we end up with a restriction leaving only a total of $O(t \cdot k + m)$ remaining variables in the restricted formula.

This completes the description of the restriction ρ. The only part that remains is to verify that ρ is natural, k-determined, and m-independent. That ρ is natural is immediate – it does not substitute any structural variable by an evaluation variable. For k-determinedness, note that for a fixed input α at most k selector circuits output 1, and thus for every gate u the value of $\mathsf{Out}_\alpha(u)$ as a function of Y can be computed by a function over those k variables. Finally, each assignment to the remaining structure variables Y gives a valid circuit and thus ρ is m-independent. □

We are ready to prove the degree lower bound, restated here for convenience.

Theorem 4.3. *For all $\varepsilon > 0$ there is a $d = d(\varepsilon)$ such that the following holds. For $n \in \mathbb{N}$, all $s \geq n^d$ and any Boolean function $f : \{0,1\}^n \to \{0,1\}$ on n bits, SoS requires degree $\Omega_\varepsilon(s^{1-\varepsilon})$ to refute* $\mathrm{Circuit}_s(f)$.

Proof. Let $G = (U, V, E)$ be an explicit bipartite graph as in Theorem 2.3, with $U = \{0,1\}^n$, $k = O_\gamma\big((n \log r)^{1+1/\gamma}\big)$, and $|V| \leq k^2 r^{1+\gamma}$ for parameters $\gamma > 0$ and $r \leq 2^n$ to be fixed later. Apply Lemma 5.1 with $t = 2^n$ along with $C_f = \perp$ to obtain, for $s \geq m \cdot \mathsf{poly}(n, k)$, a natural m-independent k-determined variable substitution ρ for $\mathrm{Circuit}_s(f)$ such that $g^{\mathrm{out}}_{s,\alpha}(Y) = \oplus_{i \in N(\alpha)} y_i$. In words, the circuit of the restricted formula on input α computes an *xor* of the neighborhood of the vertex α of G.

Apply Lemma 3.4 to ρ to conclude that if there is an SoS refutation of $\mathrm{Circuit}_s(f)\big|_\rho$ of degree d, then there is a degree $d \cdot k$ SoS refutation of the system of polynomial equations computing

$$\mathcal{P}_G = \Big\{ \bigoplus_{i \in N(\alpha)} y_i = f(\alpha) : \alpha \in \{0,1\}^n \Big\} \cup \{y_i^2 = y_i \mid i \in [m]\} \ .$$

As the graph G is a strong expander, we can apply Theorem 2.8 to get an SoS degree lower bound of $\Omega(r)$ for the XOR-CSP instance $\mathcal{P}_G$ defined over G, which in turn gives us an $\Omega(r/k)$ degree lower bound for the $\mathrm{Circuit}_s(f)\big|_\rho$ formula and hence also for the unrestricted formula.

Let us fix the parameters. We want to choose r as large as possible. However, the larger we choose r, the larger m may become, since Theorem 2.3 only

guarantees that $m \leq k^2 r^{1+\gamma}$. Let us analyze how large r can be chosen in terms of n and s.

Note that $k = \mathsf{poly}_\gamma(n)$, where we use that $r \leq 2^n$, and we write $\mathsf{poly}_\gamma(n)$ to denote some polynomial in n whose degree and coefficients may depend on γ. Hence we may choose

$$m = \frac{s}{\mathsf{poly}_\gamma(n)} \ , \tag{36}$$

according to the requirement on s in Lemma 5.1. From the guarantees of Theorem 2.3 we know that $r \geq (m/k^2)^{1/(1+\gamma)}$. Substituting m according to the previous equation we get that

$$r \geq \left(\frac{s}{k^2 \mathsf{poly}_\gamma(n)}\right)^{\frac{1}{1+\gamma}} = \frac{s^{1/(1+\gamma)}}{\mathsf{poly}_\gamma(n)} \ . \tag{37}$$

Hence if we choose γ small enough so that $\frac{1}{1+\gamma} > 1 - \varepsilon/2$ and then require s to be large enough such that the final $\mathsf{poly}_\gamma(n)$ is at most $s^{\varepsilon/2}$, we obtain the claimed lower bound. □

In the following we prove the claimed size lower bound.

Theorem 4.4. *For all $\varepsilon > 0$ there is a $d = d(\varepsilon)$ such that the following holds. Let $n \in N$ and $s \in \mathbb{N}$ such that $s \geq n^d$. If $t \geq s$ and $f \in \mathcal{F}_n(s/2, t)$, then it holds that SoS requires size $\exp\left(\Omega_\varepsilon(s^{2-\varepsilon}/t)\right)$ to refute $\mathrm{Circuit}_s(f)$.*

Proof. Apply Lemma 5.1 with the graphs from Theorem 2.3 as in the proof of Theorem 4.3. We get a natural m-independent k-determined variable substitution ρ and the formula $\mathrm{Circuit}_s(f)\big|_\rho$ over $O(t \cdot k + m)$ variables. To this formula we then apply Lemma 3.4 to obtain a degree lower bound of $\Omega(r/k)$, akin to the proof of Theorem 4.3. By setting the parameters as in the aforementioned proof we get the same degree lower bound of $\Omega_\varepsilon(s^{1-\varepsilon/3})$ for the formula $\mathrm{Circuit}_s(f)\big|_\rho$. As this formula is over few variables we can apply Theorem 2.9 to obtain an SoS size lower bound of $\exp\left(\Omega_\varepsilon\big((s^{1-\varepsilon/3} - 3k)^2/(t \cdot k + m)\big)\right)$ for the restricted formula. As variable substitutions may only decrease the size of a refutation, the same lower bound also holds for the unrestricted formula. We obtain the desired lower bound by choosing s large enough such that $s^{\varepsilon/3} \geq k = \mathsf{poly}_\varepsilon(n)$ and by recalling that $t \geq s \geq m$. □

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